In [ ]:
```
EXPERIMENT - 7
HANDLING OUTLIERS & MISSING VALUES IN A DATASET

PIN Number:`241UDS0103`

### Aim
To preprocess an environmental sensor dataset by removing duplicate rows,
detecting and removing numerical outliers using the IQR method, and imputing
missing values using Pandas and Scikit-learn SimpleImputer.
```

In [ ]:
```
## Step 1: Import the Basic Libraries
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
```

In [ ]:
```
## Step 2: Load and Analyze the Dataset
```

In [2]:
```
# Load the environmental sensor dataset
df_241UDS0103 =
pd.read_csv("Experiment_7_241UDS0103_Environmental_Sensor_Data.csv")

# Dataset information
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 123 entries, 0 to 122
Data columns (total 9 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Site_Type            121 non-null    str
 1   Temperature_C        120 non-null    float64
 2   Humidity_Percent     119 non-null    float64
 3   PM25_ug_m3           120 non-null    float64
 4   Noise_dB             121 non-null    float64
 5   Wind_Speed_kmh       123 non-null    float64
 6   Green_Cover_Percent  120 non-null    float64
 7   Water_pH             121 non-null    float64
 8   Daily_Visitors       121 non-null    float64
dtypes: float64(8), str(1)
memory usage: 8.8 KB
```

In [3]:
```
# Display the first five records
df_241UDS0103.head()
```

Out[3]: Columns 1–8 of 10

	Site_Type	Temperature_C	Humidity_Percent	PM25_ug_m3	Noise_dB	Wind_Speed_kmh	Green_Cover_Percent
0	Industrial	22.4	82.3	20.8	60.8	17.8	65.5
1	Industrial	32.8	59.5	20.9	42.3	4.7	47.8
2	Inland	23.1	39.4	8.0	62.7	18.7	24.3
3	Urban	NaN	75.3	34.2	55.8	10.8	54.2
4	Coastal	26.7	50.0	15.9	59.1	16.8	16.1

Columns 9–10 of 10 · first column repeated

	Water_pH	Daily_Visitors
0	7.00	419.0
1	7.53	481.0
2	7.55	76.0
3	6.80	165.0
4	6.75	221.0

In [4]:
```
# Check the shape of the dataset
df_241UDS0103.shape
```

Out[4]:
```
(123, 9)
```

In [ ]:
```
## Step 3: Find & Remove the Duplicates
```

In [5]:
```
# Check the number of duplicate rows
df_241UDS0103.duplicated().sum()
```

Out[5]:
```
np.int64(3)
```

In [6]:
```
# Remove duplicate rows
df_241UDS0103 = df_241UDS0103.drop_duplicates().reset_index(drop=True)

df_241UDS0103.shape
```

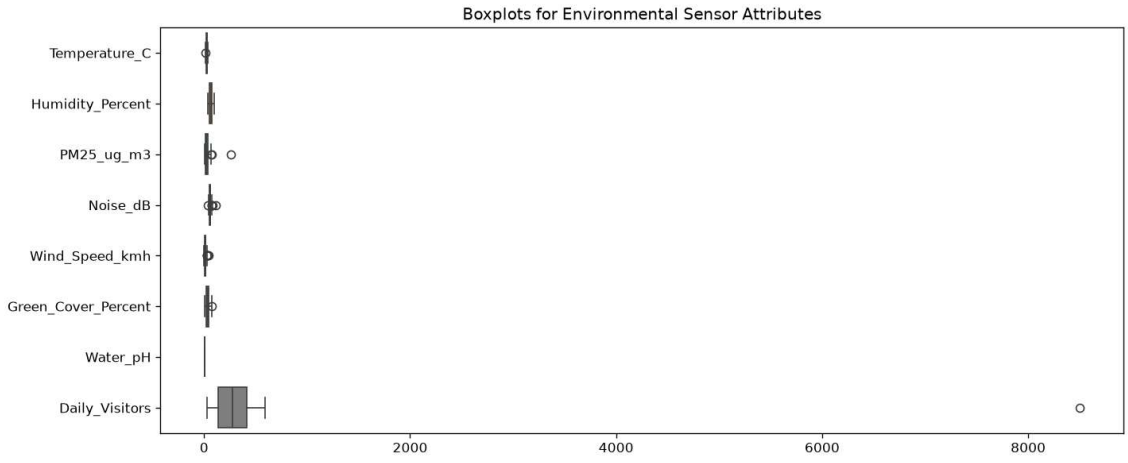
Out[6]:
```
(120, 9)
```

In [ ]:
```
## Step 4: Identify & Remove the Outliers
```

In [7]:
```
# Select numerical columns for outlier analysis
columns_241UDS0103 =
df_241UDS0103.select_dtypes(include=np.number).columns.tolist()
columns_241UDS0103
```

Out[7]:
```
['Temperature_C',
 'Humidity_Percent',
 'PM25_ug_m3',
 'Noise_dB',
 'Wind_Speed_kmh',
 'Green_Cover_Percent',
 'Water_pH',
 'Daily_Visitors']
```

In [8]:
```
# Visualize numerical columns with boxplots
plt.figure(figsize=(12, 5), dpi=120)
sns.boxplot(data=df_241UDS0103[columns_241UDS0103], orient="h")
plt.title("Boxplots for Environmental Sensor Attributes")
plt.tight_layout()
plt.show()
```



In [9]:
```
# Calculate the IQR ranges Q1 and Q3
Q1_241UDS0103 = df_241UDS0103[columns_241UDS0103].quantile(0.25)
Q3_241UDS0103 = df_241UDS0103[columns_241UDS0103].quantile(0.75)

Q1_241UDS0103, Q3_241UDS0103
```

Out[9]:
```
(Temperature_C           23.500
 Humidity_Percent        55.425
 PM25_ug_m3              14.000
 Noise_dB                54.200
 Wind_Speed_kmh           6.000
 Green_Cover_Percent     24.800
 Water_pH                 6.800
 Daily_Visitors         135.250
 Name: 0.25, dtype: float64,
 Temperature_C           29.700
 Humidity_Percent        72.050
 PM25_ug_m3              35.200
 Noise_dB                62.700
 Wind_Speed_kmh          14.075
 Green_Cover_Percent     44.400
 Water_pH                 7.380
 Daily_Visitors         417.500
 Name: 0.75, dtype: float64)
```

In [10]:
```
# Calculate IQR
IQR_241UDS0103 = Q3_241UDS0103 - Q1_241UDS0103
IQR_241UDS0103
```

Out[10]:
```
Temperature_C            6.200
Humidity_Percent        16.625
PM25_ug_m3              21.200
Noise_dB                 8.500
Wind_Speed_kmh           8.075
Green_Cover_Percent     19.600
Water_pH                 0.580
Daily_Visitors         282.250
dtype: float64
```

In [11]:
```
# Find outliers using IQR range
outliers_241UDS0103 = (
    (df_241UDS0103[columns_241UDS0103] < (Q1_241UDS0103 - 1.5 *
IQR_241UDS0103)) |
    (df_241UDS0103[columns_241UDS0103] > (Q3_241UDS0103 + 1.5 *
IQR_241UDS0103))
)

outliers_241UDS0103.sum()
```

Out[11]:
```
Temperature_C          1
Humidity_Percent       0
PM25_ug_m3             3
Noise_dB               6
Wind_Speed_kmh         7
Green_Cover_Percent    1
Water_pH               0
Daily_Visitors         1
dtype: int64
```

In [12]:
```
# Remove rows containing at least one numerical outlier
df_241UDS0103 =
df_241UDS0103.loc[~outliers_241UDS0103.any(axis=1)].reset_index(drop=True)

df_241UDS0103.shape
```

Out[12]:
```
(101, 9)
```

In [ ]:
```
## Step 5: Fill the Missing Values
```

In [13]:
```
# Find missing values in each column
df_241UDS0103.isnull().sum()
```

Out[13]:
```
Site_Type              1
Temperature_C          3
Humidity_Percent       3
PM25_ug_m3             2
Noise_dB               2
Wind_Speed_kmh         0
Green_Cover_Percent    3
Water_pH               2
Daily_Visitors         2
dtype: int64
```

In [14]:
```
# Import and create a median imputer for numerical columns
numerical_imputer_241UDS0103 = SimpleImputer(strategy="median")

numeric_columns_241UDS0103 =
df_241UDS0103.select_dtypes(include=np.number).columns.tolist()
df_241UDS0103[numeric_columns_241UDS0103] =
numerical_imputer_241UDS0103.fit_transform(
    df_241UDS0103[numeric_columns_241UDS0103]
)

# Check missing values in numerical columns
df_241UDS0103[numeric_columns_241UDS0103].isnull().sum()
```

Out[14]:
```
Temperature_C          0
Humidity_Percent       0
PM25_ug_m3             0
Noise_dB               0
Wind_Speed_kmh         0
Green_Cover_Percent    0
Water_pH               0
Daily_Visitors         0
dtype: int64
```

In [ ]:
```
### Fill categorical values using the most frequent value
```

In [15]:
```
# Create a categorical imputer
categorical_imputer_241UDS0103 = SimpleImputer(strategy="most_frequent")

df_241UDS0103[["Site_Type"]] = categorical_imputer_241UDS0103.fit_transform(
    df_241UDS0103[["Site_Type"]]
)

# Check category frequencies after imputation
df_241UDS0103["Site_Type"].value_counts()
```

Out[15]:
```
Site_Type
Rural         25
Inland        19
Urban         18
Coastal       14
Hill          14
Industrial    11
Name: count, dtype: int64
```

In [ ]:
```
## Final Step: Verify the Cleaned Dataset
```

In [16]:
```
# Display cleaned dataset information
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 101 entries, 0 to 100
Data columns (total 9 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Site_Type            101 non-null    str    
 1   Temperature_C        101 non-null    float64
 2   Humidity_Percent     101 non-null    float64
 3   PM25_ug_m3           101 non-null    float64
 4   Noise_dB             101 non-null    float64
 5   Wind_Speed_kmh       101 non-null    float64
 6   Green_Cover_Percent  101 non-null    float64
 7   Water_pH             101 non-null    float64
 8   Daily_Visitors       101 non-null    float64
dtypes: float64(8), str(1)
memory usage: 7.2 KB
```

In [17]:
```
# Check all remaining missing values
df_241UDS0103.isnull().sum()
```

Out[17]:
```
Site_Type              0
Temperature_C          0
Humidity_Percent       0
PM25_ug_m3             0
Noise_dB               0
Wind_Speed_kmh         0
Green_Cover_Percent    0
Water_pH               0
Daily_Visitors         0
dtype: int64
```

In [18]:
```
# Display the first five rows of the cleaned dataset
df_241UDS0103.head()
```

Out[18]: Columns 1–8 of 10

	Site_Type	Temperature_C	Humidity_Percent	PM25_ug_m3	Noise_dB	Wind_Speed_kmh	Green_Cover_Percent
0	Industrial	22.4	82.3	20.8	60.8	17.8	65.5
1	Industrial	32.8	59.5	20.9	42.3	4.7	47.8
2	Inland	23.1	39.4	8.0	62.7	18.7	24.3
3	Urban	27.0	75.3	34.2	55.8	10.8	54.2
4	Coastal	26.7	50.0	15.9	59.1	16.8	16.1

Columns 9–10 of 10 · first column repeated

	Water_pH	Daily_Visitors
0	7.00	419.0
1	7.53	481.0
2	7.55	76.0
3	6.80	165.0
4	6.75	221.0

In [ ]:
```
## RESULT

The environmental sensor dataset was preprocessed by removing duplicate
records, detecting numerical outliers using the IQR method, and filling
missing numerical and categorical values using median and most-frequent
imputation.

**Note:** This is synthetic practice data. The number of remaining rows
depends on the outlier-removal operation and can be checked by running the
notebook.
```